11. Reverse Array Without Using Another Array

Input:

1 2 3 4 5

Output:

5 4 3 2 1

Main.java

```java
import java.util.*;
public class Main
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n =sc.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] =sc.nextInt();

        //System.out.print(arr[i]+ " ");
        }
        int start = 0;
        int end = n-1;
        while(start < end){
            int temp = arr[start];
            arr[start] = arr[end];
            arr[end] = temp;
            start++;
            end--;
        }
        for(int i=0;i<n;i++){

            System.out.print(arr[i]+ " ");
        }

```

input

```
5
1 2 3 4 5
5 4 3 2 1

...Program finished with exit code 0
Press ENTER to exit console.
```

12. Find Maximum Difference Between Two Elements

(Second element should come after first.)

Input:

2 3 10 6 4 8 1

Output:

Maximum Difference = 8

Main.java

```java
import java.util.*;
public class Main
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr = new int[n];
        for(int i =0; i < n;i++){
            arr[i] =sc.nextInt();
        }
        int min =arr[0];
        int maxdiff = arr[1]-arr[0];
        for(int i=1;i<n;i++){
            if(arr[i]-arr[0] > maxdiff){
                maxdiff = arr[i] - arr[0];
            }
            if (arr[i] < min) {
                min = arr[i];
            }
        }

    System.out.print("Maximum Difference ="+maxdiff);
    }
}
```

input

```
7
2 3 10 6 4 8 1
Maximum Difference =8

...Program finished with exit code 0
Press ENTER to exit console.
```

13. Leaders in an Array

(A leader is greater than all elements to its right.)

Input:

16 17 4 3 5 2

Output:

17 5 2

Main.java

```java
import java.util.*;
public class Main
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] arr =new int[n];
        for(int i=0;i<n;i++){
            arr[i] =sc.nextInt();
        }
        for(int i=0;i<n;i++){
            int count =0;
            for(int j=i+1;j<n;j++){
                if(arr[i]>arr[j]){
                    count++;
                }
            }
            if(count == n-i-1){
                System.out.print(arr[i]+" ");
            }
        }
    }
}
```

input

```
6
16 17 4 3 5 2
17 5 2

...Program finished with exit code 0
Press ENTER to exit console.
```

14. Find Equilibrium Index

(Index where left sum = right sum.)

Input:

1 3 5 2 2

Output:

Equilibrium Index = 2

Main.java

```java
import java.util.*;
public class Main
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n =sc.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] =sc.nextInt();
        }
        for(int i=0;i<n;i++){
            int leftsum=0;
            int rightsum =0;
            for(int j=0;j<i;j++){
                leftsum += arr[j];
            }
            for(int j=i+1;j<n;j++){
                rightsum += arr[j];
            }
            if(leftsum == rightsum){
                System.out.println("Equilibrium Index = "+ i);
                return;
            }

        }
        System.out.println("No Equilibrium Index");
```

input

```
5
1 3 5 2 2
Equilibrium Index = 2
```

15. Find Pair with Given Sum

Input:

Array: 2 7 11 15 5
Target: 20

Output:

Pair = 15 5

Main.java

```java
import java.util.*;
public class Main
{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n =sc.nextInt();
        int[] arr = new int[n];
        for(int i=0;i<n;i++){
            arr[i] =sc.nextInt();
        }
        for(int i=0;i<n;i++){
            int sum=0;
            int Target = 9;
            for(int j=1;j<n;j++){
                sum = arr[i]+arr[j];
                if(sum == Target){
                    System.out.print("Pair =" + arr[i]+" "+arr[j]);
                    return;
                }
            }
        }
    }
}
```

input

```
5
2 7 11 15 5
Pair =2 7
```

16. Find Majority Element

(Element occurring more than n/2 times.)

Input:

2 2 1 2 3 2 2

Output:

Majority Element = 2

Main.java

```java
import java.util.*;
public class Main
{
    public static void main(String[] args) {
        int[] a = {2,2,1,2,3,2,2};
        for(int i=0;i<a.length;i++){
            int count=0;
            for(int j=1;j<a.length;j++){
                if(a[i] == a[j]){
                    count++;
                }
                System.out.print("Majority Element ="+ a[i]);
                return;

            }

        }


    }
}

```

input

```
Majority Element =2

...Program finished with exit code 0
Press ENTER to exit console.
```

17. Rearrange Array in Wave Form

Input:

1 2 3 4 5 6

Output:

2 1 4 3 6 5

Main.java

```java
import java.util.*;
public class Main
{
    public static void main(String[] args) {
        int[] ar = {1,2,3,4,5,6};
        int a=0,b=0;
        for(int i=0;i<ar.length;i+=2){
            a=ar[i];
            ar[i]=ar[i+1];
            ar[i+1]=a;
        }
        for(int i=0;i<ar.length;i++){
        System.out.print(ar[i]+" ");
        }
    }
}
```

input

```
2 1 4 3 6 5

...Program finished with exit code 0
Press ENTER to exit console.
```

18. Find Longest Consecutive Increasing Sequence

Input:

1 2 3 1 2 3 4 5

Output:

Length = 5

Main.java

```java
public class Main {
    public static void main(String[] args) {

        int[] a = {1,2,3,1,2,3,4,5};

        int count = 1;
        int max = 1;

        for(int i = 0; i < a.length - 1; i++) {

            int diff = a[i + 1] - a[i];

            if(diff == 1) {
                count++;
            } else {
                count = 1;
            }

            if(count > max) {
                max = count;
            }
        }

        System.out.println("Length = " + max);
    }
}
```

input

```
Length = 5


...Program finished with exit code 0
Press ENTER to exit console.
```

19. Find Frequency of Every Element Without Using HashMap

Input:

1 2 2 3 1 4 2

Output:

1 -> 2
2 -> 3
3 -> 1
4 -> 1

Main.java

```java
import java.util.*;

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int arr[] = new int[n];
        boolean visited[] = new boolean[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        for(int i = 0; i < n; i++) {

            if(visited[i]) {
                continue;
            }

            int count = 1;

            for(int j = i + 1; j < n; j++) {

                if(arr[i] == arr[j]) {
                    count++;
                    visited[j] = true;
```

```
7
1 2 2 3 1 4 2
1 -> 2
2 -> 3
3 -> 1
4 -> 1
```

20. Find the Smallest Positive Missing Number

Input:

3 4 -1 1

Output:

2

Main.java

```java
import java.util.*;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];
        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int ans = 1;
        while(true) {
            boolean found = false;
            for(int i = 0; i < n; i++) {
                if(arr[i] == ans) {
                    found = true;
                    break;
                }
            }
            if(found) {
                ans++;
            } else {
                break;
            }
        }
        System.out.println(ans);
    }
}
```

input

```
4
3 4 -1 1
2


...Program finished with exit code 0
Press ENTER to exit console.
```